# Recitation 9

Two-Sample Tests

Solutions and Teaching Notes

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*Goal:* Compare an observed difference between two groups with the variation we would expect if their population parameters were equal.

## Review: Antetokounmpo's free throws

**1. Turn the question into a model.** The lecture handout reports the following free throws:

| Location | Made | Missed | Attempts |
|---|---|---|---|
| Home ($H$) | 34 | 10 | 44 |
| Away ($A$) | 22 | 19 | 41 |

Let $\widetilde{x}_{H,i} \sim \text{Bernoulli}(\theta_H)$ and $\widetilde{x}_{A,j} \sim \text{Bernoulli}(\theta_A)$ indicate made shots. Assume independent shots, with a constant success probability within each location. The parameters $\theta_H, \theta_A$ are fixed but unknown. For "Is his free-throw percentage *different* at home and away?", use

$$H_0 : \theta_H = \theta_A = \theta, \qquad H_1 : \theta_H \neq \theta_A.$$

This is a **two-sample** problem: both success probabilities are unknown and both are estimated from data. For Bernoulli data, equal means also imply equal distributions; this is not true for arbitrary distributions.

**2. Measure the observed difference.** Let $X_H$ and $X_A$ denote the observed home and away datasets. Then

$$m(X_H) = \frac{34}{44} \approx 0.773, \qquad m(X_A) = \frac{22}{41} \approx 0.537,$$
$$d_{\text{obs}} = m(X_H) - m(X_A) \approx 0.236.$$

The gap is **23.6 percentage points**. Is it large relative to sampling variation? The counts alone do not answer that question.

**3. Work out the variation under the null.** Before observing the data, let $\widetilde{m}_H = 44^{-1} \sum_i \widetilde{x}_{H,i}$, $\widetilde{m}_A = 41^{-1} \sum_j \widetilde{x}_{A,j}$, and $\widetilde{d} = \widetilde{m}_H - \widetilde{m}_A$. Under $H_0$, independence gives

$$\mathbb{E}_0[\widetilde{d}] = 0, \qquad \text{Var}_0(\widetilde{d}) = \theta(1-\theta)\left(\frac{1}{44} + \frac{1}{41}\right).$$

**The variances add even though the means are subtracted.** The standard error is the square root of this variance.

## Review: from a difference to a decision

**4. Estimate the null standard error and standardize.** Under $H_0$, all 85 attempts share one success probability. Estimate it by pooling the successes and attempts:

$$\theta_{\mathrm{ML},0} = \frac{34+22}{44+41} = \frac{56}{85} \approx 0.659, \qquad \mathrm{se}_{0,\mathrm{est}} = \sqrt{\theta_{\mathrm{ML},0}(1-\theta_{\mathrm{ML},0})\left(\frac{1}{44}+\frac{1}{41}\right)} \approx 0.1029.$$

Thus $z_{\mathrm{obs}} = d_{\mathrm{obs}}/\,\mathrm{se}_{0,\mathrm{est}} \approx 2.295$: the observed gap is about 2.3 estimated null standard errors above zero. For sufficiently large samples, the corresponding random standardized statistic $\widetilde{z}_{\mathrm{test}}$ is approximately $\mathcal{N}(0,1)$ under $H_0$.

**5. Choose the tail that matches the question.** Write $\Phi(u) = \mathbb{P}(\widetilde{z} \le u)$ for $\widetilde{z} \sim \mathcal{N}(0,1)$. Choose the alternative and significance level *before* inspecting outcomes.

| Alternative | Approximate $p$ value | Reject at $\alpha = 0.05$ if |
|---|---|---|
| $\theta_H > \theta_A$ | $1 - \Phi(z_{\mathrm{obs}})$ | $z_{\mathrm{obs}} \ge 1.645$ |
| $\theta_H < \theta_A$ | $\Phi(z_{\mathrm{obs}})$ | $z_{\mathrm{obs}} \le -1.645$ |
| $\theta_H \ne \theta_A$ | $2[1 - \Phi(\lvert z_{\mathrm{obs}}\rvert)]$ | $\lvert z_{\mathrm{obs}}\rvert \ge 1.960$ |

For the original question, both directions count as evidence:

$$p_{\mathrm{two}} \approx \mathbb{P}(|\widetilde{z}| \ge 2.295) \approx 0.0218 < 0.05.$$

Reject $H_0$ at the 5% level. A *prespecified* home-is-better alternative would instead give $p_{\mathrm{right}} \approx 0.0109$.

**6. Say what the result means.** Under the equal-probability model, a standardized gap at least this extreme in either direction has approximate probability 2.2%. This is evidence of a difference; it is *not* a 2.2% probability that $H_0$ is true. It does not show that crowd taunts caused the difference: location was not randomly assigned, and opponents, fatigue, and game conditions may differ.

**Check the approximation.** A common rule of thumb asks for at least 10 expected makes and 10 expected misses in each group, using the pooled estimate. Here these are $(28.99, 15.01)$ at home and $(27.01, 13.99)$ away. This supports a normal approximation, but does not establish independence. Shots within a game may be dependent. The nominal false-positive rate is *approximate* for this plug-in $z$ test and relies on the model.

**The same idea for other means.** For two independent samples of sizes $n_A, n_B$, with population means $\mu_A, \mu_B$ and *known* variances $\sigma_A^2, \sigma_B^2$,

$$\mathrm{se}(\widetilde{m}_B - \widetilde{m}_A) = \sqrt{\frac{\sigma_B^2}{n_B} + \frac{\sigma_A^2}{n_A}}, \qquad z_{\mathrm{obs}} = \frac{m(X_B) - m(X_A)}{\sqrt{\sigma_B^2/n_B + \sigma_A^2/n_A}} \quad (H_0 : \mu_B = \mu_A).$$

This is an exact Gaussian test for independent Gaussian observations with known variances; otherwise a CLT may justify an approximation. With unknown variances and small samples, a suitable two-sample $t$ procedure is needed.

## Practice problems

Three core problems; Problem 4 is a backup. A calculator is useful. Use $\Phi(1.645) \approx 0.95$, $\Phi(1.960) \approx 0.975$, and $\Phi(2.295) \approx 0.9891$.

### Problem 1: Why do we pool the free throws?

*Independent work: 8 minutes. Based on the lecture example.* Use the home/away data on page 1 and the independent Bernoulli model.

(a) Write the likelihood under $H_0 : \theta_H = \theta_A = \theta$, up to a factor independent of $\theta$. Derive $\theta_{\mathrm{ML},0}$. Why not use the unweighted average of the two observed success proportions?

(b) Starting from $\mathrm{Var}(\widetilde{x}_{H,i}) = \theta(1-\theta)$ under $H_0$, derive $\mathrm{Var}_0(\widetilde{d})$. Explain both denominators and the plus sign.

(c) Compute the estimated null SE, standardized statistic, and two-sided $p$ value. State the decisions at $\alpha = 0.05$ and $\alpha = 0.01$.

(d) A student treats $22/41$ as the *known* away success probability and includes only the home group's variance contribution in the SE. What source of uncertainty is missing?

**Solution.** **(a)** There are 56 makes and 29 misses in total, so

$$L_0(\theta) \propto \theta^{56}(1-\theta)^{29}, \qquad \ell_0'(\theta) = \frac{56}{\theta} - \frac{29}{1-\theta}.$$

Setting the derivative to zero gives $\boxed{\theta_{\mathrm{ML},0} = 56/85}$. The second derivative is negative on $(0,1)$, and the likelihood vanishes at both endpoints, so this is the maximum. Equivalently,

$$\theta_{\mathrm{ML},0} = \frac{44\, m(X_H) + 41\, m(X_A)}{85}.$$

Each attempt receives the same weight. An unweighted average would give equal weight to groups with different numbers of attempts.

**(b)** Independence within each group gives $\mathrm{Var}_0(\widetilde{m}_H) = 44\theta(1-\theta)/44^2 = \theta(1-\theta)/44$; similarly, $\mathrm{Var}_0(\widetilde{m}_A) = \theta(1-\theta)/41$. Between-group independence makes their covariance zero, hence

$$\boxed{\mathrm{Var}_0(\widetilde{d}) = \mathrm{Var}_0(\widetilde{m}_H) + \mathrm{Var}_0(\widetilde{m}_A) = \theta(1-\theta)(1/44 + 1/41).}$$

The minus sign is squared when computing the variance.

**(c)** Using unrounded fractions,

$$d_{\mathrm{obs}} = 0.23614, \quad \mathrm{se}_{0,\mathrm{est}} = 0.10291, \quad z_{\mathrm{obs}} = 2.29460, \quad p_{\mathrm{two}} \approx 0.02176.$$

Reject at 5%; do not reject at 1%. Failure to reject at 1% does not establish equal success probabilities.

**(d)** The away proportion is itself estimated from only 41 attempts. Its variance contribution $\theta(1-\theta)/41$ is missing. Using the same pooled plug-in value, the proposed SE is about 0.07147, too small; it overstates the evidence against $H_0$.

## Problem 2: Comparing treatment and control means

*Independent work: 8 minutes. Archived Exercise 11.19.* An experiment compares a control group ($A$) with an independent treatment group ($B$). Observations are approximately normally distributed, and the standard deviation of an individual measurement in either group is known to be $\sigma = 5$. Each group has 25 observations.

(a) What is the standard error of $\widetilde{m}_B - \widetilde{m}_A$?

(b) For $H_0 : \mu_B = \mu_A$ versus $H_1 : \mu_B > \mu_A$, give the level-0.05 rejection region, both in standardized units and in the original measurement units.

(c) Suppose the test's $p$ value is 0.07. Would you reject at $\alpha = 0.10$? At $\alpha = 0.05$?

(d) What is the level-0.05 rejection region if instead $H_1 : \mu_B \neq \mu_A$? Explain why the cutoff changes.

*Discussion:* Why is $5/\sqrt{50}$ not the SE in part (a)?

**Solution.** **(a)** Each sample mean has variance $5^2/25 = 1$. Independence gives

$$\boxed{\mathrm{se}(\widetilde{m}_B - \widetilde{m}_A) = \sqrt{\frac{25}{25} + \frac{25}{25}} = \sqrt{2} \approx 1.414.}$$

The value $5/\sqrt{50}$ is the SE of the grand mean of 50 independent observations with variance 25, not the difference of two group means.

**(b)** Under $H_0$, the standardized difference is approximately standard Gaussian. Let $d_{\mathrm{obs}} = m(X_B) - m(X_A)$. Reject when

$$\boxed{z_{\mathrm{obs}} = \frac{d_{\mathrm{obs}}}{\sqrt{2}} \geq 1.645} \quad \Longleftrightarrow \quad \boxed{d_{\mathrm{obs}} \geq 1.645\sqrt{2} \approx 2.326.}$$

Only a sufficiently positive difference supports this alternative.

**(c)** Since $0.07 \leq 0.10$, reject at 10%. Since $0.07 > 0.05$, do not reject at 5%. Choose the significance level in advance; changing it after seeing the $p$ value changes the planned decision rule.

**(d)** Split the 5% null tail area equally between the two directions:

$$\boxed{|z_{\mathrm{obs}}| \geq 1.960} \quad \Longleftrightarrow \quad \boxed{|d_{\mathrm{obs}}| \geq 1.960\sqrt{2} \approx 2.772.}$$

Thus reject if $d_{\mathrm{obs}} \leq -2.772$ or $d_{\mathrm{obs}} \geq 2.772$. The positive cutoff is larger because only 2.5% of the null distribution lies in the upper rejection tail.

*Connection to the free throws:* Both tests divide a difference by its null standard error. Here the variances are known. In the Bernoulli example, the unknown common success probability determines both variances and is estimated using pooled data.

### Problem 3: What does the test tell us?

*Independent work: 7 minutes. Adapted from archived Exercise 9.26(b)–(d), with extensions to the free-throw example.* Mark each statement true or false and justify your answer in one or two sentences. For a false statement, give a correct replacement.

(a) If the $p$ value is 0.03, the test rejects at significance level 0.02.

(b) If a test using the rule "reject when $p \leq \alpha$" rejects at significance level 0.06, then its $p$ value is at most 0.06.

(c) The free-throw test's two-sided $p$ value 0.0218 is the probability that the home and away success probabilities are equal.

(d) If we compute away minus home instead of home minus away, the two-sided $p$ value is unchanged.

(e) After seeing that the home percentage is higher, we may replace the planned two-sided test by a one-sided test and halve its $p$ value while keeping the advertised significance level.

(f) Rejecting equality establishes that crowd taunts caused the lower away success rate.

**Solution.** **(a) False.** $0.03 > 0.02$, so do not reject. The same result would reject at a prespecified 5% level.

**(b) True.** This is exactly the stated rejection rule. A smaller significance level might lead to a different decision.

**(c) False.** The $p$ value measures how extreme the data are under the null model. Here it is approximately the probability, under $H_0$, of a standardized difference with magnitude at least 2.295. It is not a posterior probability of $H_0$.

**(d) True.** Reversing the subtraction changes $z_{\text{obs}}$ to $-z_{\text{obs}}$ but leaves $|z_{\text{obs}}|$ unchanged. For a one-sided test, the sign and the alternative must be kept consistent: "home is better" becomes a *left*-tail test for away minus home.

**(e) False.** The direction must be chosen before examining the outcomes. If we always choose whichever direction the observed effect favors and then use a one-sided 5% cutoff, we reject for $|z_{\text{obs}}| \geq 1.645$. Under an ideal continuous standard-normal null, that event has probability 10%, not 5%.

**(f) False.** A difference in probabilities is not a causal attribution. This observational comparison does not isolate crowd taunts from other differences between home and away games. The independent-shot model may also be imperfect.

*Exit check:* Complete the sentence: "Assuming the null model, the $p$ value is the probability of _____." The missing phrase is "a test statistic at least as extreme as the one observed, in the direction(s) specified by the alternative."

### Problem 4 (backup): Is the normal approximation credible?

*Independent work: 5 minutes. Adapted from archived Recitation Worksheet 11, Problem 2.* A study records exposure to an environmental factor and disease status:

| | Not exposed | Exposed |
|---|---|---|
| No disease | 128 | 7 |
| Disease | 19 | 7 |

Model individuals as independent, with disease probabilities $\theta_N$ and $\theta_E$ in the two groups.

(a) State hypotheses for whether the disease probabilities differ, and find the observed exposed-minus-unexposed difference.

(b) Find the pooled null estimate and the expected disease/no-disease counts in each group using that estimate.

(c) Does the "at least 10 expected successes and failures per group" guideline hold? Explain why computing a normal-tail $p$ value alone would not settle whether the approximation is reliable.

**Solution.** **(a)** Test $H_0 : \theta_E = \theta_N$ against $H_1 : \theta_E \neq \theta_N$. The observed difference is

$$\boxed{d_{\text{obs}} = \frac{7}{14} - \frac{19}{147} \approx 0.37075.}$$

The difference is about 37.1 percentage points.

**(b)** The common-probability estimate is $\theta_{\text{ML},0} = 26/161 \approx 0.16149$. Multiplying each group size by this value and its complement gives

| Expected count under fitted $H_0$ | Not exposed | Exposed |
|---|---|---|
| Disease | 23.74 | 2.26 |
| No disease | 123.26 | 11.74 |

**(c)** No: the expected exposed disease count is only 2.26. A mechanical pooled $z$ calculation gives $z \approx 3.602$ and an approximate two-sided $p \approx 0.000316$, but that small number does not validate its Gaussian calibration. The guideline is a diagnostic, not a theorem asserting failure or success. A suitable exact method, such as Fisher's exact test for this $2 \times 2$ table, avoids this normal approximation; its calculation is beyond this recitation.

Even a well-calibrated test would assess association, not establish that exposure caused disease. The original archive asks for a significance test; this adaptation emphasizes checking the assumptions first.

## Instructor notes: 75-minute plan

| Time | Activity |
|---|---|
| 0–7 min | Free-throw table; two unknown probabilities; formulate a two-sided alternative before calculating. |
| 7–16 min | Derive the difference's null mean and variance; explain pooling and standardization. |
| 16–25 min | One versus two tails; $p$ value, decision, approximation checks, and the limits of a causal conclusion. |
| 25–42 min | Problem 1: 8 minutes independent work, 9 minutes discussion. |
| 42–58 min | Problem 2: 8 minutes independent work, 8 minutes discussion. |
| 58–72 min | Problem 3: 7 minutes independent work, 7 minutes discussion. |
| 72–75 min | Exit check and questions. |

*Fast-class option:* Keep all independent-work periods. Shorten discussions by 2 minutes each for Problems 1 and 2 and by 1 minute for Problem 3. Then use minutes 67–72 for Problem 4 and 72–75 for its discussion and the exit check.

**Board sequence.** Write the two proportions, then ask whether a 23.6-point gap is surprising. Build the chain

$$\text{equal probabilities} \implies \mathbb{E}_0[\widetilde{d}] = 0 \implies \text{se}_{0,\text{est}} \approx 0.103 \implies z \approx 2.295 \implies p_{\text{two}} \approx 0.022.$$

Sketch a bell curve with both tails beyond $\pm 2.295$ shaded. Ask which tail would count for a prespecified "home is better" question. Do not describe the positive raw difference alone as the two-sided evidence: its *magnitude* is what matters.

**Assumptions and notation to emphasize.** The true common parameter $\theta$ is deterministic and unknown; $\theta_{\text{ML},0} = 56/85$ is its observed estimate under $H_0$. The random means have tildes, and $X_H, X_A$ denote observed datasets. The pooled null SE is appropriate for testing equal Bernoulli probabilities. An unrestricted confidence interval for their difference commonly uses separate group estimates instead; do not conflate the two calculations. Independent groups do not by themselves ensure independence of shots *within* groups. A game-level dependence model would require more information than these two totals.

**Sources and adaptations.** The review and Problem 1 follow Carlos Fernandez-Granda's *Two-Sample Tests* handout, especially the free-throw counts and the two tail probabilities. We call the square-root quantity a standard error (the slide labels it "variance") and describe Gaussian calibration as approximate. Problem 2 preserves Exercise 11.19 from the Fall 2025 archive's `mine/r10/main.tex`; it also appears in archived Worksheet 9. Problem 3(a)–(c) adapts Exercise 9.26(b)–(d) from `mine/r8/main.tex`; (d)–(f) connect these ideas to the lecture example. Problem 4 preserves the counts in archived Worksheet 11, Problem 2, and changes the task to checking normal-approximation conditions. The archived worksheets themselves carry Spring 2025 headers.